



The Universe by Powers of Ten: From a Picnic to the Edge of Everything

【【生词双高亮】BBC科普精选：宇宙到底有多大？ | 从野餐毯到宇宙边缘】

Summary: Powers of Ten: A Journey Through Scale and Time. This video pays homage to the iconic 1977 film by Charles and Ray Eames, taking viewers on an exponential voyage from a picnic in Sicily to the edge of the observable universe. It illustrates the scale of the cosmos, from human dimensions to interstellar space, highlighting key astronomical milestones. The journey showcases advancements in space exploration, such as the Voyager missions and the Hubble Telescope, and concludes with the cosmic microwave background radiation—the oldest light in the universe. The film invites reflection on how our understanding of the universe may evolve by 2077.

摘要：《十的次方：穿越尺度和时间的旅程》。本视频致敬了1977年查尔斯和雷·伊姆斯的经典电影，带领观众从西西里岛的野餐地指数级飞跃到可观测宇宙的边缘。它展示了从人类尺度到星际空间的宇宙规模，并突出了关键的天文学里程碑。这段旅程呈现了太空探索的进展，如旅行者号任务和哈勃望远镜，并以宇宙微波背景辐射——宇宙中最古老的光——作为结尾。影片引发人们思考：到2077年，我们对宇宙的认知将如何演变。





双语字幕 | 生词高亮 | 智能文稿

🕒 Estimated Reading Time: 10 min.

📖 四级生词 📖 六级生词 📖 雅思生词 📖 托福生词 📖 专八生词 📖 SAT生词 📖 考研生词 📖
GRE生词

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▶ Trying to comprehend how big the universe is is one of those questions that astrophysicists grapple with all the time

试图#理解#宇宙的浩瀚是天体物理学家们一直在#努力应对#的问题之一

▶ Trying to explain it in a way that's easy to understand

尝试用通俗易懂的方式解释它

▶ Well, that's a whole other challenge

这则完全是另一个挑战

▶ In 1977, Charles and Ray Eames, hugely influential American designers, released one of the most elegant and creative pieces of science communication of modern times

1977年，极具影响力的美国设计师查尔斯和雷·伊姆斯发布了当代最优雅且富有创意的科学传播作品之一

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▶ Powers of ten.

《十的次方》

▶ It took the viewers on a journey from a picnic blanket near Lake Michigan to the edge of the known universe and back again

它带领观众从密歇根湖边的#野餐#毯出发，直达已知宇宙的边缘再返回

▶ Over 40 years later as a humble homage to this groundbreaking film

四十多年后，作为对这部开创性影片的#谦逊#致敬#

▶ We're going to take a similar journey through time and space and see how our understanding has changed along the way

我们将展开类似的时空之旅，见证人类认知如何随时间演进

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▶ As in 1977, we'll start with a picnic though this time we're on the island of Sicily in Italy rather than Lake Michigan

如同1977年，我们将从#野餐#开始——不过这次是在意大利西西里岛而非密歇根湖畔

▶ We'll start with a scene one metre wide viewed from one metre away and every ten seconds

初始视野是一米见方的场景，观测距离为一米，之后每十秒

▶ We're going to move out to ten times further away so the scene will be ten times wider

观测距离将扩大十倍，视野范围也随之扩大十倍

▶ This square is ten metres wide and in ten seconds the next square will be ten times that

这个方块宽十米，十秒后下一个方块将是其十倍

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▶ the movement may seem linear

这种移动看似#线性#

▶ But we're actually accelerating exponentially into the distance

但实际上我们正在呈指数级加速远离

▶ This square is a hundred metres wide, the distance someone can run in ten seconds

这个方块宽百米，相当于人类十秒内能奔跑的距离

▶ Well, if they're running at 43 kilometres an hour just under the speed of your cat, one kilometre

按时速43公里计算（略低于家猫的奔跑速度），可达一公里

▶ Though our picnics are indistinguishable now, we can still clearly see the impact of human activity on the world

虽然野餐场景已#难以辨认#，但仍能清晰看到人类活动对世界的影响

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▶ Ten thousand metres are ten to the four metres, this is the distance of supersonic plane travels in ten seconds

万米即十的四次方米，这是#超音速#飞机十秒内飞行的距离

▶ And we're now reaching the highest altitude flown by such a plane, ten to the five metres

此时达到此类飞机的最高#飞行高度#——十的五次方米

▶ This is the distance the International Space Station travels in ten seconds

这是国际空间站十秒内航行的距离

▶ There it goes, from here on, human activity will be lost to sight, we're at the scale of countries

看，从这里开始，人类活动将消失于视野，我们进入国家尺度

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▶ One million metres, or ten to the six metres, we've long left Earth's atmosphere

百万米（十的六次方米），我们早已脱离地球大气层

▶ And soon we'll see the whole planet, what a duel

很快将看到整个星球，多么壮观的对决

▶ Ten million metres

千万米量级

▶ The invisible magnetosphere shields us from the dangerous ionising radiation of space

#无形的#磁层保护我们免受太空危险电离#辐射#的侵害

▶ Ten to the eight metres, this line extends at the speed of light

十的八次方米处，这条线以光速延伸

▶ This is the time it takes for light to reach us from the Moon's orbit, the age of Moonlight

这是月光从月球#轨道#抵达地球所需的时间——月光年龄

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▶ The Earth is now just a pale blue dot in the sky full of stars

地球此刻只是繁星苍穹中一个淡蓝圆点

▶ Even as we accelerate away, the stars appear stationary because they're so much further away

即便持续#加速#远离，恒星依然显得#静止#——因为它们遥远得多

▶ So much empty space, let's illustrate the orbits of the planets in our solar system, otherwise this could get a bit dull

如此多虚空，让我们标注太阳系行星轨道以免枯燥

▶ Here comes the orbit of Venus, N Mars, and now Mercury

这是#金星#轨道、#火星#轨道，以及#水星#轨道

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▶ Since 2010 the NASA Solar Dynamics Observatory has been using an extreme ultraviolet filter to monitor the activity of our Sun

自2010年起，NASA太阳#动力学#天文台一直使用极端#紫外#滤镜监测太阳活动

▶ Finally, we reach the orbit of the outer planets, the gas giants, but just specks at this distance

最终抵达外行星#轨道#——气态巨行星在此距离仅是微尘

▶ There's the orbit of Pluto, one of the dwarf planets of the Kuiper belt, ten to the thirteen metres, and we're moving out of the solar system

这是柯伊伯带矮行星冥王星的轨道（十的十三次方米），我们正飞出太阳系

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▶ In 2012 the Voyager One spacecraft became the first human artifact to make this journey, followed in 2018 by its twin of Voyager Two

2012年旅行者一号#探测器#成为完成此旅程的首个人类#造物#，其姊妹探测器旅行者二号于2018年紧随其后

▶ Both were launched in 1977, the year the Eames were working on powers of ten

两者均发射于1977年——正是伊姆斯夫妇创作《十的次方》之年

▶ We're heading into interstellar space, our Sun is just one of billions of stars, and still at this distance the night sky looks very similar to what we see at home

我们正进入星际空间，太阳仅是数十亿恒星之一，而此距离的夜空与地球所见仍极为相似

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▶ This square is ten to the sixteen metres, the distance light travels in one year, one light year

这个方块宽十的十六次方米，即光一年传播的距离——一光年

▶ Here's our closest neighbour Alpha Centauri, three stars for the price of one, with planets orbiting around them

这是近邻比邻星——三合星系统附带环绕行星

▶ I'd love to know what's going on there

真想知道那里正在发生什么

▶ Thanks to data collected by the Gaia spacecraft, we're building a detailed 3D map of the Milky Way

借助盖亚#探测器#收集的数据，我们正在绘制详细的#银河#系三维地图

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▶ There are between 100 and 400 billion stars in our galaxy alone, and clouds of dust and gas like these nebula, where new stars are born

仅银河系就有1000亿到4000亿颗恒星，还有孕育新恒星的星云尘埃气体

▶ Images sent from the Hubble Space Telescope have been blowing our minds for a generation

哈勃太空#望远镜#传回的图像震撼了整整一代人

▶ As we move away, we begin to see the great flat spiral of our galaxy

继续远离时，我们开始看到银河系巨大的扁平#螺旋#结构

▶ A few hundred billion stars rotating around a black hole, Sagittarius A star, 4.2 million times more massive than our Sun

数千亿颗恒星环绕着比太阳重420万倍的黑洞——人马座A*运行

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▶ We now think supermassive black holes reside at the centre of nearly all galaxies

如今我们认为几乎所有星系中心都#存在#超大质量黑洞

🎧 These two dwarf galaxies are the Magellanic clouds, which together with at least 80 others, make up what's known as the local group of galaxies

这两个矮星系是麦哲伦云，它们与至少80个其他星系共同组成了所谓的本星系群

🎧 Ten to the 22, one million light years, soon we'll pass the super giant elliptical galaxy M87
10的22次方米，一百万光年，我们将很快掠过超巨型#椭圆# #星系#M87

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🎧 And if we switch to radio waves, we can glimpse the supermassive black hole at its centre
若切换至无线电波，我们便能#窥见#其中心的超大质量黑洞

🎧 Switching back to visible light as we traverse the Virgo Supercluster, each tiny dot not a star, but a galaxy

当我们#穿越#室女座超星系团时切回可见光，每个微小光点都不是恒星而是#星系#

🎧 Billions of stars floating in an ever-growing void, ten to the 24 metres, the limits of our vision in 1977

数十亿恒星漂浮在日益膨胀的#虚空#中，10的24次方米是1977年人类的观测极限

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🎧 But over 40 years later, we can show a bit more

但四十余年后的今天，我们的视野已更加深远

🎧 Clusters of galaxies arranged along filaments, like the Pisces Cedar Supercluster Complex

星系团沿纤维状结构排列，如双鱼-#鲸鱼#座超星系团复合体

🎧 At 10 to the 26 metres, we switch our view to microwave

在10的26次方米尺度，我们将视野切换至#微波#

🎧 And we can now see the current limit of our vision

此刻映入眼帘的是当前的观测极限

▶ This light forms a sort of wall all around us

这道光芒在我们周围形成某种壁垒

▶ The light and dark patches show differences in temperature by fractions of a degree

明暗斑块显示着百分之几度的温度差异

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▶ Revealing where matter was beginning to clump together to form the first galaxies, shortly after the Big Bang

揭示了大爆炸#后不久物质开始#聚集#形成首批星系的区域

▶ This light is known as the Cosmic Microwave Background Radiation

这道光芒被称为#宇宙# #微波#背景#辐射#

▶ Ten to the 27 metres, one followed by 27 zeros

10的27次方米，1后面跟着27个零

▶ Beyond this point, the nature of the universe is truly uncharted and debated

超越此界限后，宇宙的本质便真正进入未知且存在争议的领域

▶ This light was emitted around 380,000 years after the Big Bang

这道光是在大爆炸#后约38万年时发出的

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▶ Before this time, the universe was so hot that it was not transparent to light

在此之前的宇宙温度极高，光线无法穿透

▶ Is there simply more universe out there yet to be revealed?

是否还存在更多尚未被发现的宇宙？

▶ Or is this region still expanding, generating more universe?

或是这片区域仍在扩张，不断生成新宇宙？

🎵 Or even other universes with different physical properties to our own?

甚至存在物理特性与我们宇宙截然不同的其他宇宙？

🎵 How will our understanding of the universe have changed by 2077?

到2077年，人类对宇宙的认知将发生怎样的变革？

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🎵 How many more powers of ten are out there?

还有多少十次方尺度等待探索？

🎵 From a picnic blanket unscissly to the very edge of our understanding, I salute the eems

从#野餐毯#的方寸之间到认知的最前沿，我#致敬#eems团队

🎵 For the way they told this beautiful story, the story of the universe

因他们讲述这个美丽故事——宇宙史诗的独特方式

